Theorem 18.1

If G is a finite abelian group then G is isomorphic to a direct product of cyclic groups whose orders are powers of primes:

$$G \cong \mathbb{Z}_{p_1^{r_1}} \times \mathbb{Z}_{p_2^{r_2}} \times \ldots \times \mathbb{Z}_{p_k^{r_k}}$$

for primes $p_1, \ldots, p_k$ (not necessarily distinct) and integers $r_1, \ldots, r_k \geq 1$ such that $p_1^{r_1} \cdot p_2^{r_2} \cdot \ldots \cdot p_k^{r_k} = |G|$.

Theorem 18.2

Let G be an abelian group, and let H, K be subgroups of G such that

1) $H \cap K = \{e\}$
2) for every $g \in G$ there exist $h \in H$, $k \in K$ such that $g = hk$

Then $G \cong H \times K$.

Theorem 18.3

Let G be a finite abelian group. Assume that $|G| = p^r m$ where p is a prime, $r \geq 1$ and m is a number which is not divisible by p. Then $G \cong H \times K$ where $|H| = p^r$ and $|K| = m$.

Corollary 18.4

If G is a finite abelian group and $|G| = p_1^{r_1} p_2^{r_2} \cdot \ldots \cdot p_k^{r_k}$ where $p_1, p_2, \ldots, p_k$ are distinct primes then

$$G = G_1 \times G_2 \times \ldots \times G_k$$

where $|G_i| = p_i^{r_i}$.

Theorem 18.5

If G is an abelian group such that $|G| = p^n$ for some prime p and let a be the element of the largest order in G. Then there exists a subgroup $K \subseteq G$ such that

$$G \cong \langle a \rangle \times K \cong \mathbb{Z}_{p^r} \times K$$

Corollary 18.6

If G is an abelian group such that $|G| = p^n$ for some prime p then G is a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p^{k_1}} \times \mathbb{Z}_{p^{k_2}} \times \ldots \times \mathbb{Z}_{p^{k_m}}$$

for some $k_1, k_2, \ldots, k_m$.